End-Member Mixing Analysis of Winyah Bay and Santee River Plumes

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ABSTRACT

This study develops and applies an observational end-member mixing analysis (EMMA) to separate adjacent river plumes on the South Atlantic Bight, focusing on Winyah Bay (WB) and the Santee River (SR), whose outlets are <15 km apart. Shipboard hydrographic and optical data were collected during March 9th-13th, 2025 aboard the R/V *Savannah* at 63 CTD stations along three cross-shelf transects spanning the region between the WB and SR outflows. Plume-layer properties were defined using a density-based plume-depth estimate and used to evaluate candidate tracers for distinguishing these freshwater sources. Among CTD-recorded variables, salinity and colored dissolved organic matter (CDOM) averaged within the plume layer exhibited the most linear, near-conservative behavior and were selected as tracers for a three-endmember (WB, SR, ocean) mixing model. Endmembers were defined from representative near-field stations and fractions were obtained by solving linear mixing equations for salinity and CDOM. Observations outside the mixing region defined by the three endmember tracer values were projected to the nearest point on the mixing boundary to quantify model mismatch. Results show that traditional plume visualizations (profiles, contours, maps, and scatterplots) could distinguish plume-influenced from oceanic waters but could not reliably separate WB from SR-influenced waters due to strong overlap in property space. In contrast, EMMA produced spatially coherent source fractions: stations along the southern transect clustered toward the SR and ocean edge of ternary space, while northern transect stations clustered toward the WB and ocean edge. Source-dominance maps varied systematically with wind forcing, consistent with downwelling-favorable winds promoting along-shelf advection and upwelling-favorable winds promoting northeastward advection and increased SR influence across much of the study region. Model mismatch was generally small, with one pronounced outlier excluded. These results demonstrate that a simple, widely available tracer pair (salinity and CDOM) can provide quantitative separation of neighboring plumes from in situ measurements, offering a practical tool for interpreting observations and evaluating models in multi-plume coastal systems.

1. Introduction

Where rivers and estuaries enter the ocean, a buoyant freshwater plume forms that transports sediments, nutrients, and pollutants onto the coastal shelf (Beusen et al. 2005; Hickey et al. 2010). These plumes strongly control coastal stratification and circulation by freshening the surface layer, modifying vertical mixing, and driving along-shelf and cross-shelf flows (Fong and Geyer 2001; Hetland and MacDonald 2008; Horner-Devine et al. 2009, 2015; MacDonald and Geyer 2004; Yankovsky et al. 2022). They also influence the transport of river-borne nutrients and organic matter, which can enhance primary production, but also increase oxygen demand, degrade water quality, and contribute to coastal hypoxia (Chen et al. 2021; Fennel et al. 2006; Friedland et al. 2025; Li et al. 2024; Ortega et al. 2025; Rabalais et al. 2002). Because these physical and biogeochemical impacts depend on the way river and ocean waters mix, understanding how plumes evolve within coastal waters is important to predict how land-based inputs affect coastal ecosystems.

River plumes are characterized by sharp gradients in salinity and optical properties such as turbidity and colored dissolved organic matter (CDOM) (Horner-Devine et al. 2015). Being less dense than ambient seawater, plumes float and can extend tens of kilometers from their source (Warrick et al. 2007). During their lifetime, plumes mix and dilute with surrounding waters, while their direction and spreading rate are shaped by physical drivers, including river discharge, winds, tides, Coriolis, and coastline geometry (Horner-Devine et al. 2009). Depending on the balance of these forces, a plume may remain closely attached to the coast or detach and propagate offshore (Horner-Devine et al. 2009). Upwelling-favorable winds can drive offshore spreading and the formation of cross-shelf plumes (Yankovsky et al. 2022), while downwelling-favorable winds tend to compress plumes against the coast (Dykstra et al. 2024; Horner-Devine et al. 2009).

There are multiple established methods to track river plumes. Satellites with sensors for true color, infrared, radar, and altimetry give great spatial resolution and are widely used (Devlin et al. 2015; Fiedler and Laurs 1990; Frey and Osadchiev 2021; Horner-Devine et al. 2009; Thomas and Weatherbee 2006). Satellites can view the ocean in different wave lengths, which can be used to differentiate between plume waters from the surrounding ocean. In addition, satellite altimetry can detect the slight sea-surface elevation due to plume buoyancy. However, satellites provide only surface information, their revisit times can be limiting, and cloud cover often obscures the ocean surface.

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In-situ measurements are also commonly used to track river plume movement, including salinity, temperature, turbidity, dissolved oxygen, CDOM, chlorophyll, and mixing rates (Burrage et al. 2002; Del Castillo et al. 2001; Kastner et al. 2023; Warrick et al. 2007; Yankovsky et al. 2022). These measurements allow for the tracking and characterization of the plumes, as well as the measurement of the sediments, nutrients, and pollutants that they transport (Warrick et al. 2007). However, these measurements can be costly and often require access to specialized research vessels and equipment.

Although many individual plumes have been studied, interactions between neighboring plumes remain relatively understudied (Gong et al. 2019; Warrick and Farnsworth 2017). When two plumes occupy the same coastal region, their waters can converge, overlap, or mix depending on wind and other forcings (Warrick and Farnsworth 2017). These interactions can complicate the identification of freshwater sources and can alter the distribution of sediments and nutrients, resulting in consequences for regional productivity (Gong et al. 2019). Most prior work on multi-plume systems emphasizes modeling as the main method of study. However, observational methods are needed that can distinguish adjacent plume contributions using field data.

Winyah Bay (WB) and the Santee River (SR) both discharge freshwater from central South Carolina to the South Atlantic Bight through outlets less than 15 km apart. Winyah Bay receives freshwater input from the Pee Dee, Waccamaw, Black, and Sampit rivers with an estimated average total freshwater input into the estuary of 557 m^3 s^{-1} (Kim and Voulgaris 2008; Patchineelam 1999). Inland, the discharge of the Santee River is regulated by a series of hydroelectric dams, so the coastal discharge reflects both river runoff and dam operations (Hockensmith 2004). Near the coast, the Santee River then bifurcates into two channels that flow into the coastal region through outlets less than 3 km apart. Data collected from 1986-2002 showed that the Santee River had an average stream flow of 311 m^3 s^{-1} (Hockensmith 2004). Although Winyah Bay supplies the larger freshwater input, the Santee delivers roughly half as much on average, and together these two systems provide a substantial amount of freshwater to the coastal ocean in this region.

This study develops and applies an observational method to separate neighboring river plumes on the South Atlantic Bight. Specifically, this study aims to: (1) develop a method that uses commonly measured water parameters to calculate water-mass fractions from Winyah Bay, the Santee River and the ocean; (2) apply the method to in-situ data collected in and around the Winyah Bay and Santee River plumes to classify individual stations as Winyah-dominated, Santee-dominated, or oceanic; and (3) evaluate the consistency and validity of these classifications by comparing the results with measured winds and MODIS ocean-color imagery.

2. Methods

Field data were collected aboard the R/V Savannah from March 9th-13th, 2025, off the coast of South Carolina between the Winyah Bay and Santee River outflows. The study region spans approximately 32.9°-33.3° N and 78.9°-79.3° W and lies on a broad and shallow continental shelf.

A total of 63 stations were sampled along three transects (A, B, and C) oriented perpendicular to the shoreline to capture the cross-shelf structure of the freshwater plumes (Figure 1). Transect A was located just north of the Winyah Bay jetties, and transect C was positioned just south of the mouth of the Santee River, while transect B was placed between the mouths where plume interaction was expected. The stations along each transect were visited multiple times during the five-day cruise: transect B was occupied four times, transect A three times, and transect C twice. The station names follow the convention of transect letter (A, B, or C) followed by the station number, a period, and the visit number (e.g., the third visit to station 7 along transect A is labeled A7.3). This sampling strategy

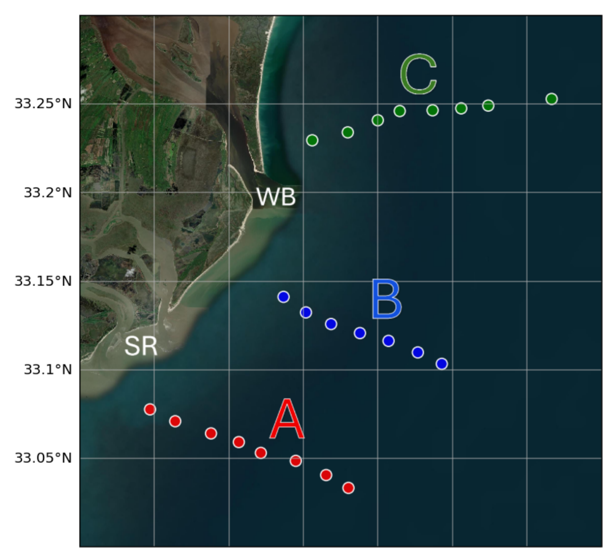

Fig. 1. Map of the March 2025 study area and CTD transects. Transect station locations are shown, with the Winyah Bay outlet and the Santee River marked for reference. Basemap imagery is ArcGIS World Imagery.

was designed to capture the temporal evolution of the plumes as well as the interactions between them.

Vertical profiles were collected at each station using a Sea-Bird SBE 11plus CTD system sampling at 24 Hz. The parameters recorded by the CTD include temperature, salinity, and pressure, as well as photosynthetically active radiation (PAR), chlorophyll fluorescence, dissolved oxygen, CDOM fluorescence, and nitrogen saturation. In addition to station profiles, continuous near-surface measurements were obtained from a flow through system that draws seawater from the ship's underway. Surface salinity and temperature from the flow-through system were recorded every 6 seconds. Wind speed and direction were measured using port and starboard anemometers mounted on the ship and the Shipboard Computer System (SCS) corrected measured winds to true wind speed and direction using the ship's speed over ground. Wind data were also logged every 6 seconds.

a. Processing and Analysis

The CTD profiles were processed using Sea-Bird Electronics software following the manufacturer's recommended procedures. Processing steps included a low-pass filter and a loop edit to remove excess noisy data associated with the heave of the ship and to remove the surface soaking period before the internal pump was active.

Plume depth (D) at each station was estimated following Arneborg et al. (2007) as outlined by Papageorgiou (2023):

$$D = \frac{2\int_{-h}^{0} \frac{\rho(z)-\rho_0}{\rho_0} z \, dz}{\int_{-h}^{0} \frac{\rho(z)-\rho_0}{\rho_0} \, dz}$$

where z is the elevation below the sea surface, h is the total depth of the water column, $\rho(z)$ is the density at elevation z, and ρ_0 is the reference ambient density. In this equation, the plume depth D is based on how density changes with depth compared to a background reference density ρ_0. The computation was implemented numerically using the discrete depth and density values of each cast. The shallowest observed density in each profile was assumed to extend to the surface, and the reference density ρ_0 was defined as the highest density observed in the profile. The vertical integrals in the plume depth equation were computed using a trapezoidal approximation. This procedure produced an estimated plume depth for each station, and then the plume-layer properties were calculated by averaging all measurements between the shallowest recorded sample and the estimated plume depth.

To investigate potential differences between the Winyah Bay-influenced and Santee River-influenced regions, a series of exploratory visualizations were produced to evaluate whether the stations of transects A and C exhibited distinct water-mass characteristics. Analyses examined salinity, temperature, density, CDOM fluorescence, chlorophyll fluorescence, dissolved oxygen, nitrogen saturation, and photosynthetically active radiation (PAR), using both full-depth CTD profiles and plume-layer averages. These variables were compared across transects by evaluating their vertical structure, cross-shelf patterns, and relationships among properties, and these comparisons guided the interpretation of plume structure and informed whether simple property thresholds could help distinguish between source waters. The detailed patterns revealed by these plots are presented in the Results section.

The selection of variables to effectively distinguish between freshwater sources followed the criteria outlined by Cuoco et al. (2021): tracers should exhibit an approximately linear relationship, thus indicative of physical mixing; tracers should behave conservatively over the spatial and temporal scales of interest so that differences between samples primarily reflect physical mixing rather than in situ sources or sinks; endmember sources must be significantly distinct in at least one tracer to make source contributions identifiable; and endmember tracer values should be reasonably consistent across the investigated area and sampling period to avoid bias from drifting source values over time. All CTD-derived variables were considered potential tracers, but properties commonly influenced by air-sea exchange or biological processes (e.g., temperature, chlorophyll fluorescence, dissolved oxygen) were expected to be less reliable unless they appeared stable across the study region. Tracers were therefore evaluated using plume-layer averages, with scatterplots and linear regressions used to assess linearity, and notched box-plots and Wilcoxon rank-sum tests used to assess whether endmember sources were separable in at least one property. Based on these criteria and the prior use of salinity and CDOM as conservative tracers in estuarine and plume systems (Bai et al. 2013; Chen and Hu 2017; Wei et al. 2023; Zhong et al. 2025), plume-layer salinity (S) and CDOM fluorescence (C) were selected as tracers for a three-endmember mixing model.

All data were collected outside of the source waters due to operational constraints; therefore, the endmember values for salinity and CDOM were calculated from the plume-layer averages at three stations selected to best represent the distinct source waters in the study region. The freshwater endmembers were chosen from the most river-influenced (lowest salinity) portion of each transect, while avoiding "extreme" observations that departed from the linear salinity-CDOM relationship used to characterize mixing. Station C2.2 was selected as the Winyah Bay (WB) endmember and station A4.3 as the Santee River (SR) endmember, based on their positions as the best low-salinity candidates within transects C and A, as well as their geographic proximity to their representative freshwater outflows. To reduce the likelihood of selecting stations already influenced by cross-source mixing, endmember selection also considered wind-driven plume behavior. Under sustained downwelling-favorable or low-wind conditions, buoyant plumes can be advected down the coast (Dykstra et al. 2024; Horner-Devine et al. 2009), in this case, causing the WB plume to overlap the SR outflow. The ocean (OC) endmember was defined by station B7.3, selected as one of the most offshore, highest-salinity, least stratified stations. Its full-depth salinity profile showed a vertically uniform, high-salinity water column consistent with minimal plume influence. B7.3 also combined near-zero CDOM with a salinity value representative of ocean water, avoiding an overly saline, zero CDOM endmember choice that would compress the mixing model and reduce the precision of the results. Together, these stations form the three endmembers in salinity-CDOM space, providing the endmember values used in the mixing analysis below.

b. End-member mixing analysis

This end-member mixing analysis (EMMA), adapted from Christophersen and Hooper (1992); Han et al. (2012); Cuoco et al. (2021), was used to estimate the fractional contributions of water from three sources: Winyah Bay (WB), Santee River (SR), and the ocean (OC). Using the selected tracers (S and C), the three-endmember system was formulated as:

$$\begin{aligned} f_{\mathrm{WB}}\, S_{\mathrm{WB}} + f_{\mathrm{SR}}\, S_{\mathrm{SR}} + f_{\mathrm{OC}}\, S_{\mathrm{OC}} &= S_{\mathrm{obs}} \\ f_{\mathrm{WB}}\, C_{\mathrm{WB}} + f_{\mathrm{SR}}\, C_{\mathrm{SR}} + f_{\mathrm{OC}}\, C_{\mathrm{OC}} &= C_{\mathrm{obs}} \\ f_{\mathrm{WB}} + f_{\mathrm{SR}} + f_{\mathrm{OC}} &= 1 \end{aligned}$$

where S_{obs} and C_{obs} are the observed plume-layer salinity and CDOM at a given station, while S_i and C_i are the tracer values for endmember $i \in \{\mathrm{WB}, \mathrm{SR}, \mathrm{OC}\}$ and f_i are the corresponding water-mass fractions. For each station observation $(S_{\mathrm{obs}}, C_{\mathrm{obs}})$, the endmember fractions $\mathbf{F} = [f_{\mathrm{WB}}, f_{\mathrm{SR}}, f_{\mathrm{OC}}]$ were solved using the linear system:

$$\begin{bmatrix} S_{\mathrm{WB}} & S_{\mathrm{SR}} & S_{\mathrm{OC}} \\ C_{\mathrm{WB}} & C_{\mathrm{SR}} & C_{\mathrm{OC}} \\ 1 & 1 & 1 \end{bmatrix} \mathbf{F} = \begin{bmatrix} S_{\mathrm{obs}} \\ C_{\mathrm{obs}} \\ 1 \end{bmatrix}.$$

The solution provides the three endmember fractions that correspond to coordinates within a ternary space. Importantly, the estimated fractions can be negative, these are stations outside of the mixing triangle in salinity-CDOM space. For these stations, the solution is constrained to the triangle by drawing a straight line from the observed station toward the geometric center of the endmember triangle and then finding the point where that line intersects the triangle boundary. The distance between the observed station point and the intersection point in salinity-CDOM space defines the error magnitude.

3. Results

Meteorological conditions varied considerably during the five-day cruise. A wind-vector timeseries

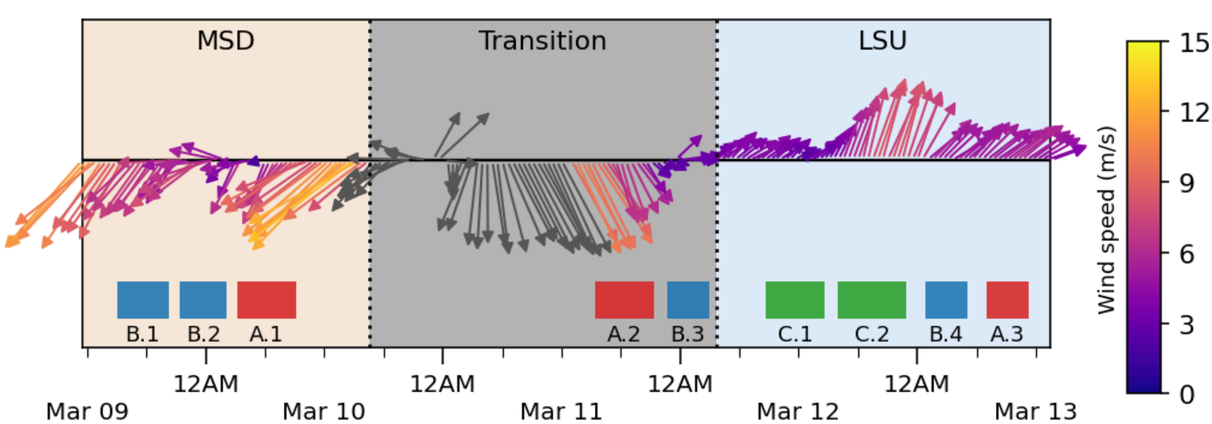

FIG. 2. Time series of wind vectors spanning the cruise (March 9-13, 2025). Arrow direction shows wind direction (relative to true north) and arrow magnitude indicates relative wind speed while arrow color shows actual wind speed (m/s). A data gap filled with NDBC Frying Pan Shoals buoy winds is shown in gray. Wind-regime periods (MSD, Transition, LSU) are marked and transect sampling windows are marked along the timeline.

from the shipboard anemometers (Figure 2) shows three distinct wind regimes during the cruise, in order: Moderately-Strong Downwelling-favorable (MSD), Strong Transition, and Low-Strength Upwelling-favorable (LSU). During MSD, winds blew generally to the southwest, which is aligned along the shelf and therefore downwelling-favorable for this region. The mean wind speeds were 7.60 m s^{-1} with maximum wind speeds reaching 12.92 m s^{-1}. During the Strong Transition period, the winds rotated from SW to SE, with a mean speed of 8.36 m s^{-1} and maximum wind speeds reaching 14.83 m s^{-1}. During LSU, the winds blew consistently to the NE and were much weaker than during the previous regimes. The mean speeds were 5.14 m s^{-1}, with maximum wind speeds reaching only 8.56 m s^{-1}.

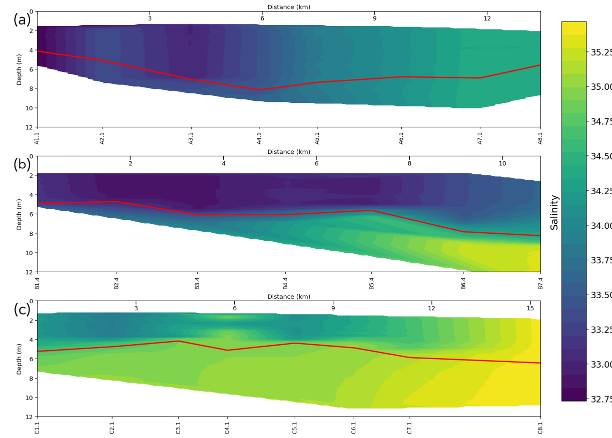

FIG. 3. Salinity cross-sections for three selected transects (A.1, B.4, and C.1). Station locations are indicated along the transect axis, and the calculated plume depth at each station is overlaid as a red line.

a. Visual analysis

Salinity contours by transect visit (Figure 3) show a clear difference in vertical structure between the MSD and LSU regimes. During MSD, the water column was vertically mixed, leading to weak vertical salinity gradients as shown by transect A.1 (Figure 3a). During LSU, clear stratification developed, with a fresher layer overlying the denser and more saline layers, leading to sharper vertical salinity gradients as shown by transects B.4 and C.1 (Figure 3b and 3c). CDOM contours closely mirrored the salinity structure, while other property contours were not useful to distinguish even a plume layer.

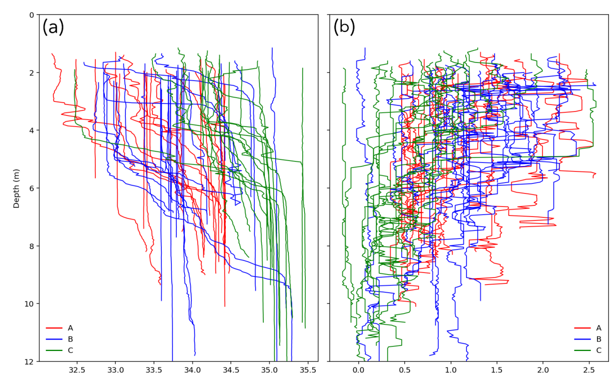

FIG. 4. Depth profiles for all stations colored by transect. (a) Salinity profiles. (b) CDOM profiles.

Plume-layer salinity varied among the transects but showed only moderate differences overall. Depth profiles (Figure 4a) and notched box-plots of plume-layer averages (Figure 5a) indicate that transects A and B exhibited similar, relatively fresh plume-layer salinities, whereas transect C was more saline. Wilcoxon rank-sum tests on plume-layer averaged salinity confirmed that A and B did not differ significantly, while C had significantly higher salinity than both A and B (Bonferroni-adjusted $p < 0.01$). Maps of plume-layer salinity (Figure 6) showed high values near the coast and in the plume region and lower values offshore, especially along transect C.

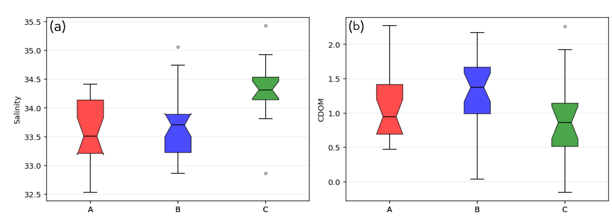

FIG. 5. Notched box-plots of plume-layer averaged station values by transect. (a) Salinity. (b) CDOM. Boxes are colored by transect, with notches indicating a 95% confidence interval around the median. Pairwise Wilcoxon rank-sum tests indicate higher salinity in transect C ($p < 0.01$) and higher CDOM in transect B relative to C ($p = 0.0137$); CDOM in transect A was not significantly different from either transect.

Plume-layer CDOM showed a weaker contrast between transects. Depth profiles (Figure 4b) and notched box-plots of plume-layer averages (Figure 5b) showed that transect B tended toward higher CDOM and transect C toward lower values, while transect A overlapped both. Wilcoxon rank-sum tests on plume-layer averages by station showed only the B and C differences were statistically significant (Bonferroni-adjusted $p = 0.0137$). Maps of plume-layer CDOM largely mirrored the salinity patterns.

To evaluate whether simple property combinations could distinguish between Winyah Bay-influenced and Santee River-influenced waters, a series of scatter plots were constructed using plume-layer averages (Figure 7). The salinity-CDOM plot exhibited the clearest structure of all scatterplots. However, data from transects A, B, and C overlapped strongly and only weak grouping by transect was evident. In a 3 dimensional scatter plot of salinity, CDOM, and chlorophyll points again showed heavy overlap among transects. Any slight separation could be explained by the salinity-CDOM relationship alone. Chlorophyll did not provide additional power for separating the water masses.

Depth profiles, contour plots, maps, and scatter plots revealed broad differences between plume-influenced waters and more oceanic waters. However, these methods could not clearly distinguish Winyah Bay-influenced waters from Santee River-influenced waters because the data between transects A, B, and C overlapped strongly. Although salinity and CDOM appeared to show the most distinction between transects, no single property or combination of properties provided a clear distinction between waters. These limitations motivated the application of a three-end-member mixing analysis using salinity and CDOM to quantify the relative contributions of Winyah Bay, the Santee River, and oceanic waters.

b. End-member mixing analysis

The End-member mixing analysis (EMMA) was applied to plume-layer averaged salinity and CDOM to estimate the fractional contributions of water from Winyah Bay (WB), the Santee River (SR) and the ocean (OC). These estimated fractions are visualized on a ternary plot (Figure 8). The stations of each transect are plotted mostly in accordance with their geographic region, with few exceptions. The C stations are clustered along the WB-OC side of the triangle, and the A stations are concentrated along the SR-OC side.

Stations were classified as Winyah, Santee or ocean-dominant by assigning each station to the end member with the largest fraction. The resulting spatial patterns are shown in the geographic classification maps grouped by wind regime (Figure 9). During the MSD regime (Figure 9a), landward stations were generally Winyah-dominant, whereas more seaward stations tended to be Santee or ocean-dominant. During the Transition period (Figure 9b), transect B stations were predominantly Winyah-dominant, while transect A shifted from Santee-dominant nearshore stations to ocean-dominant offshore stations. Satellite imagery acquired on March 11th approximately one hour before sampling (Small 2025) indicated plume waters advecting to the southwest and a pronounced tidal

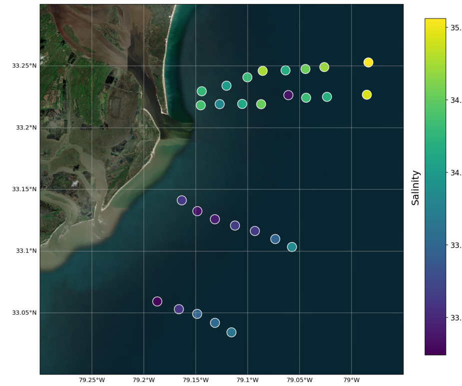

FIG. 6. Map of stations during LSU period with points colored by plume-layer averaged salinity.

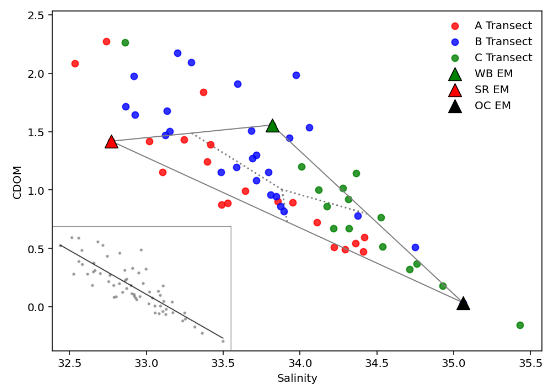

FIG. 7. Plume-layer averaged salinity versus CDOM for all stations colored by transect. Endmember stations and the endmember mixing triangle are shown, with interior dotted lines indicating dominant-endmember regions. A linear regression for all stations is shown in the bottom left (r = -0.84, p <0.001).

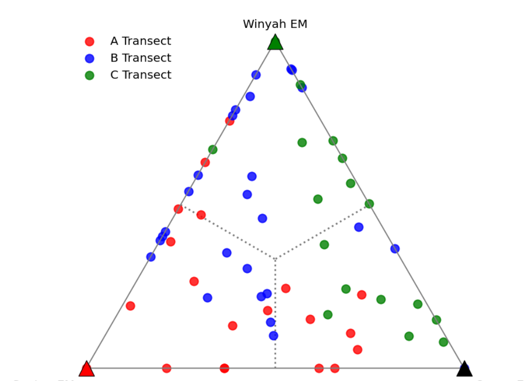

FIG. 8. Ternary (endmember fraction) plot for all stations colored by transect. Station positions reflect EMMA-derived fractions of the three endmembers, with endmembers at the vertices. Dotted lines indicate dominant-endmember regions within the ternary space.

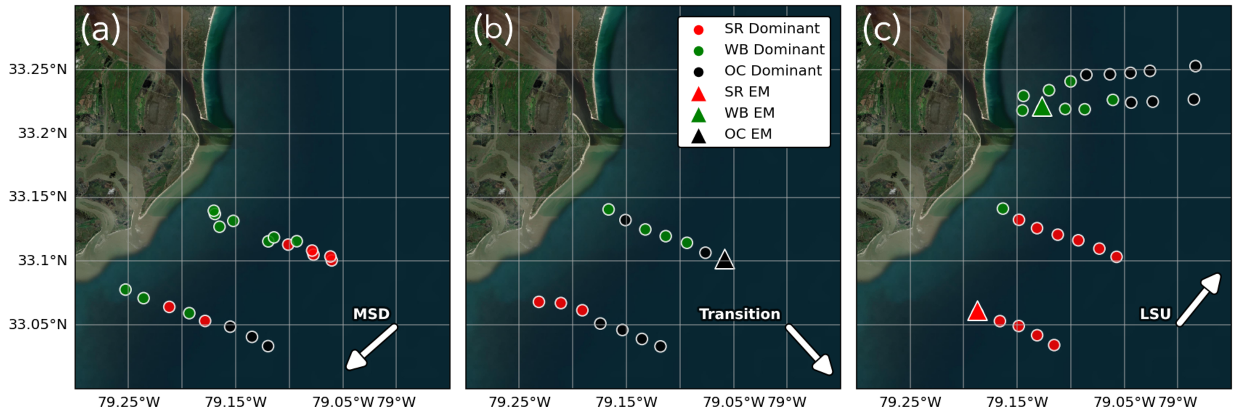

FIG. 9. Transect maps grouped by wind regime: (a) MSD, (b) Transition, and (c) LSU. Stations are colored by dominant endmember and endmember stations are marked. An arrow on each map shows the mean wind direction during that regime.

pulse expanding from Winyah Bay. During the LSU period (Figure 9c), transect C stations transitioned from Winyah-dominant nearshore to Ocean-dominant offshore while transects A and B were nearly entirely Santee-dominant. Satellite imagery collected on March 12th between the first and second samplings of transect C (Small 2025) showed plume waters advecting to the northeast, along with a Winyah Bay tidal pulse that expanded and was advected in the same direction.

Figure 10 summarizes the mixing-model mismatch for plume-layer salinity-CDOM observations relative to the endmember mixing triangle. In salinity-CDOM space (Figure 10a), most stations are captured within the triangle and therefore have zero mismatch, indicating that their tracer values can be represented as a mixture of the three end members. Stations that fall outside of the triangle have a nonzero mismatch quantified as the shortest distance to the triangle boundary. This histogram (Figure 10b) shows that all but two of these stations with nonzero mismatch exhibit error magnitudes of less than 1. One such station, A1.2, is a very pronounced outlier relative to the rest of the data set and was excluded from subsequent analyses.

4. Discussion

This study developed and applied an end-member mixing analysis to separate neighboring river plumes. By using a three-end-member mixing model based on salinity and CDOM, this method quantified fractional contributions from Winyah Bay, the Santee River, and the ocean. In contrast to traditional plume analyses based on individual properties, the EMMA results produced water-mass fractions that are intuitive and consistent with changing conditions. Several pieces of evidence suggest that the EMMA method captured the dominant patterns of plume composition during the cruise. First, ternary plots showed that stations grouped in ways that match their geographic setting: C stations tended to lie along the Winyah-ocean side of the triangle and A stations along the Santee-ocean side.

Second, the classification maps are consistent with the observed wind regimes. During MSD and the Transition period, the EMMA indicated mostly Winyah-dominated waters near the coast, which is consistent with downwelling-favorable winds driving plumes down the coast (Dykstra et al. 2024; Horner-Devine et al. 2009), and further accompanied by satellite imagery (Small 2025) showing plume waters advecting to the southwest. During LSU, the EMMA indicated mostly Santee-dominated waters in transects A and B, which is consistent with upwelling-favorable winds driving plumes away from the coast (Yankovsky et al. 2022), and again further accompanied by satellite imagery (Small 2025) showing plume waters now advecting to the northeast.

Third, the error metric provides a check on the fit of the model with the in-situ values measured at each station. Most stations exhibited small error magnitudes clustered near zero (Figure 10a), indicating that the three-endmember mixing model explained the observed plume-layer salinity and CDOM well. Only one station had a notably large error magnitude, corresponding to extremely fresh, high-CDOM water relative to the rest of the observed stations. This cast likely sampled a new tidal pulse of Santee water, while neighboring stations still reflected older, partially mixed plume waters.

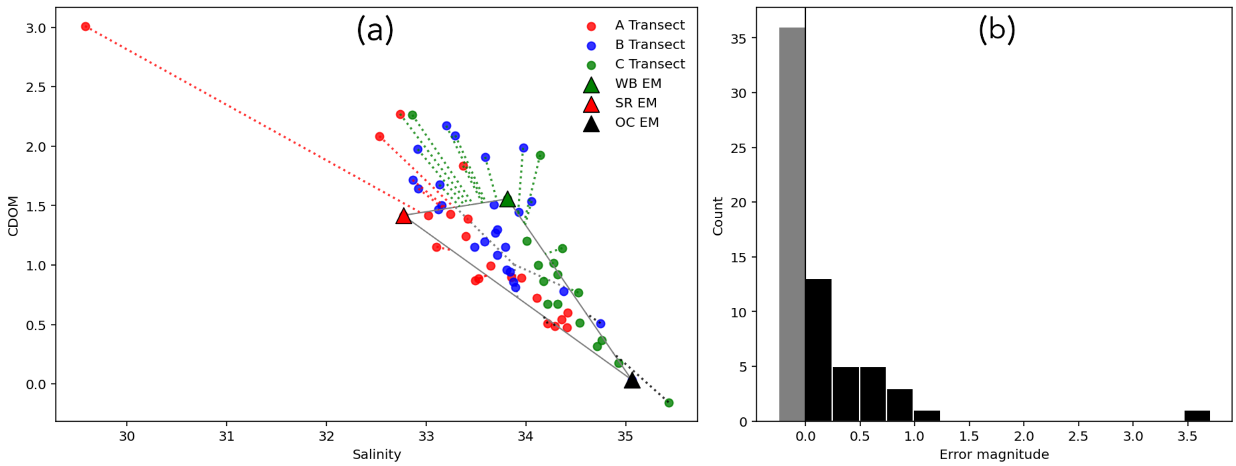

Fig. 10. Mixing-model mismatch and station error relative to the three-endmember triangle. (a) Plume-layer averaged salinity versus CDOM for all stations with the end-member mixing triangle outlined and endmember stations marked. Points are colored by transect. Dotted lines show each station's mismatch to the mixing model (distance to the triangle boundary) and are colored by the station's dominant endmember. Dotted gray lines inside the triangle show the boundaries between endmember dominated regions. (b) Histogram of mismatch magnitudes: black bars show stations with nonzero mismatch (outside the triangle), and the gray bar represents stations captured by the mixing model (zero mismatch).

a. Motivation for an End-Member Mixing Analysis

Exploratory analyses based on depth profiles, contour plots, maps, notched box-plots, and cross-property scatter plots showed that, generally, these methods and properties were not enough to cleanly separate Winyah-influenced from Santee-influenced waters. However, these simple methods did show that salinity and CDOM were the most promising tracers for an EMMA.

Depth profiles (Figure 4) and contour plots (Figure 3) clearly differentiated plume-influenced waters from more oceanic waters, but transects A and C did not show visual differences in these plots. Notched box-plots (Figure 5) and two-dimensional scatter plots (Figure 7) only revealed moderate differences among transects, and the addition of chlorophyll did not improve the separation of water sources. Many stations values lie in overlapping data regions where simple criteria such as single variable thresholds could not distinguish whether a measured station was more strongly influenced by Winyah Bay or the Santee River.

The EMMA method addresses these issues by using salinity and CDOM together with mass-balance equations involving each end member. The three-endmember mixing model then turns these overlapping property values into quantitative water-mass fractions. The resulting ternary diagrams and classification maps provide a broad scale picture of Winyah, Santee, and ocean influence that could not be obtained from previous analyses. Viewing the classifications in map form (Figure 9) emphasizes spatial patterns of overall plume structure, such as during different wind regimes, rather than focusing on ambiguous values from individual stations.

b. Limitations

The EMMA method relies on several assumptions and limitations. A primary limitation is the assumption that tracers for each freshwater end member are effectively constant during the study period. The generally small misfits and consistent ternary and spatial patterns suggest that this assumption is reasonable over the five-day cruise. However, it may be challenged under rapidly changing hydrological conditions.

For example, a storm during Day 2 likely increased the freshwater discharge from the Santee and Winyah with a delay of a couple of days. If discharge and tracer concentrations evolve substantially between fixed endmember characterization and plume sampling, an EMMA may bias endmember fractions by pulling fresher stations toward the endmember whose tracer values best match the shifted plume properties. Future work should address this by repeatedly sampling end members through time or across changing hydrologic conditions.

This method is also sensitive to endmember characterization, and in this dataset there is a tradeoff between selecting stations that are representative of plume conditions during the study period and stations that are closest to unmixed source waters. Initial tests using the freshest available station (e.g., the initial Transect A choice of A1.2) produced unrealistic fractions because that station was an outlier in the salinity-CDOM space, skewing the solution toward Winyah and the ocean at the expense of the Santee. Using a more saline but more representative station improved the ternary patterns, illustrating that end members must be stable and representative of the conditions being modeled. However, because end members were inferred near plume origins rather than sampled within the estuaries, they likely already incorporate early mixing and transformation with ambient shelf waters, reducing their representativeness of the true source end members. Future applications of this method should address this by designing targeted end-member sampling within each estuary and pairing it with near-field plume sampling to quantify how end-member choice affects resulting fractions.

This method is best interpreted on a scale of regional patterns rather than individual stations. A small number of stations display fractions that are difficult to interpret in isolation. However, when combined with transects and maps, any anomalies do not undermine the overall patterns inferred from the EMMA results for the Winyah Bay and Santee River system.

c. Previous work

Most prior work on interacting river plumes has emphasized numerical modeling. Idealized simulations by Warrick and Farnsworth (2017) demonstrated that along coasts with many small watersheds, collisions and coalescence of neighboring plumes should be common. Their model showed a variety of collision types controlled by differences in plume density, and that once plumes overlap, salinity fields alone can be insufficient to distinguish the source of plume waters. They argued that in regions where multiple small plumes coalesce, attributing a plume boundary to a particular river requires additional tracers beyond only salinity. This study provides an observational example of that argument. By using salinity in combination with CDOM, the EMMA method offers a way to distinguish plume waters where salinity alone cannot. In this way, this method uses commonly measured water properties to meet the need for independent tracers in multi-plume settings.

More complex modeling, such as the ROMS simulations in the Pearl River Delta (Gong et al. 2019), shows how plume interactions can reshape circulation and material transport. The study by Gong et al. (2019) used passive tracers in models to separate contributions from different river outlets and to test different mechanisms such as plume-driven jets, suppression of upstream spreading, and modified estuarine circulation. The EMMA classifications from this study could be used in a complementary role: to provide observational estimates of source-water fractions that could be used to initialize, evaluate, or improve models in real multi-plume systems.

Finally, this study extends previous applications of three-end-member mixing models in a new context. Cuoco et al. (2021) used an EMMA framework in a groundwater system to quantify contributions from three aquifers. Han et al. (2012) applied a similar approach to the Pearl River plume system, where endmembers represented the Pear River plume, offshore ambient surface water, and subsurface upwelled water. Han et al. (2012) focused on separating physical mixing from biological nutrient utilization. In contrast, the endmembers of this study represent two neighboring rivers and the ambient ocean water. The EMMA method is similar to the previous works, but rather than separating river and upwelled waters or aquifers contributions to groundwater, this analysis distinguishes among multiple river sources that are optically and hydrographically similar.

d. Future work

Although this is a specific case study on the Winyah Bay and Santee River system, this method is well suited to coastal regions where several small to moderate sized rivers with comparable discharge rates influence the same coastal region and their plumes can merge and interact. Previous studies have identified conditions where coalescing plumes may be important for coastal biogeochemistry and pollutant transport. Examples include Hawke's Bay in New Zealand, parts of the northeast Black Sea, segments of central Chile, and portions of the southern California coast where multiple rivers discharge within tens of kilometers of each other. In these environments, separating the contributions of neighboring plumes is important to track the inputs of nutrients and pollutants and their effects on coastal ecosystems (Collins and Macdonald 2025; Osadchiev and Sedakov 2019; Saldías et al. 2012; Warrick and Farnsworth 2017).

Applying this framework elsewhere would require conservative or near-conservative tracers that differ among the source waters, representative end-member sampling that captures the source properties over the timescale of interest, and a sampling design that covers the spatial extent of the overlapping plumes. In some coastal settings, salinity and CDOM may be insufficient. Additional tracers may be necessary, such as temperature, nutrients, or dissolved ions (e.g., Ca, Mg, Na, Cl^-, SiO_2), assuming that they meet the tracer criteria outlined in Cuoco et al. (2021), exhibit a linear relationship, conservative behavior over the region, clear contrasts among sources, and reasonably stable end-member values.

The EMMA framework can be extended further. With more tracers, it can accommodate more than three end members, allowing for the separation of additional rivers or the inclusion of upwelled waters. In future applications, the same mixing equations could be applied to individual depth bins instead of only to plume layer averages. This would give vertical profiles of the Winyah, Santee, and ocean fractions, allowing visualizations of the vertical plume structures across transects.

In conclusion, the EMMA method presented here demonstrates that multi-plume systems can be separated into source fractions using commonly measured water properties. Applied to the Winyah Bay and Santee River system, it provided spatially consistent estimates of the relative influence of Winyah Bay, the Santee River, and the ambient ocean, where simple property analyses could not. This study highlights that the EMMA method can be useful for interpreting coastal observations in regions where river plume coalescence is likely and for linking measurements to sediment and nutrient budgets. When combined with physical and biogeochemical observations, EMMA could provide a bridge between satellite imagery, which maps surface plume extent, and numerical models, which simulate plume dynamics and interactions, and together offer a practical method for future observational studies in other multi-plume coastal environments.

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References

Arneborg, L., V. Fiekas, L. Umlauf, and H. Burchard, 2007: Gravity Current Dynamics and Entrainment—A Process Study Based on Observations in the Arkona Basin. *Journal of Physical Oceanography*, **37 (8)**, 2094–2113, doi:10.1175/JPO3110.1, URL http://journals.ametsoc.org/doi/10.1175/JPO3110.1.

Bai, Y., D. Pan, W. Cai, X. He, D. Wang, B. Tao, and Q. Zhu, 2013: Remote sensing of salinity from satellite-derived CDOM in the Changjiang River dominated East China Sea. *Journal of Geophysical Research: Oceans*, **118 (1)**, 227–243, doi:10.1029/2012JC008467, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2012JC008467.

Beusen, A. H. W., A. L. M. Dekkers, A. F. Bouwman, W. Ludwig, and J. Harrison, 2005: Estimation of global river transport of sediments and associated particulate C, N, and P. *Global Biogeochemical Cycles*, **19 (4)**, 2005GB002 453, doi:10.1029/2005GB002453, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2005GB002453.

Burrage, D. M., M. L. Heron, J. M. Hacker, T. C. Stieglitz, C. R. Steinberg, and A. Prytz, 2002: Evolution and dynamics of tropical river plumes in the Great Barrier Reef: An integrated remote sensing and in situ study. *Journal of Geophysical Research: Oceans*, **107 (C12)**, doi:10.1029/2001JC001024, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2001JC001024.

Chen, C., G. Gong, K. Chiang, F. Shiah, C. Chung, and C. Hung, 2021: Scaling effects of a eutrophic river plume on organic carbon consumption. *Limnology and Oceanography*, **66 (5)**, 1867–1881, doi:10.1002/lno.11729, URL https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.11729.

Chen, S., and C. Hu, 2017: Estimating sea surface salinity in the northern Gulf of Mexico from satellite ocean color measurements. *Remote Sensing of Environment*, **201**, 115–132, doi:10.1016/j.rse.2017.09.004, URL https://linkinghub.elsevier.com/retrieve/pii/S0034425717304169.

Christophersen, N., and R. P. Hooper, 1992: Multivariate analysis of stream water chemical data: The use of principal components analysis for the end-member mixing problem. *Water Resources Research*, **28 (1)**, 99–107, doi:10.1029/91WR02518, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91WR02518.

Collins, C., and H. S. Macdonald, 2025: Modelling the variability and dynamics of river plumes in Hawke's Bay, Aotearoa New Zealand. *Frontiers in Marine Science*, **Volume 12 - 2025**, URL https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2025.1536550.

Cuoco, E., S. Viaroli, T. H. Darrah, V. Paolucci, R. Mazza, and D. Tedesco, 2021: A geometrical method for quantifying end-members' fractions in three-component groundwater mixing. *Hydrological Processes*, **35 (11)**, e14 409, doi:10.1002/hyp.14409, URL https://onlinelibrary.wiley.com/doi/10.1002/hyp.14409.

Del Castillo, C. E., P. G. Coble, R. N. Conmy, F. E. Müller-Karger, L. Vanderbloemen, and G. A. Vargo, 2001: Multispectral in situ measurements of organic matter and chlorophyll fluorescence in seawater: Documenting the intrusion of the Mississippi River plume in the West Florida Shelf. *Limnology and Oceanography*, **46 (7)**, 1836–1843, doi:10.4319/lo.2001.46.7.1836, URL https://aslopubs.onlinelibrary.wiley.com/doi/10.4319/lo.2001.46.7.1836.

Devlin, M., C. Petus, E. Da Silva, D. Tracey, N. Wolff, J. Waterhouse, and J. Brodie, 2015: Water Quality and River Plume Monitoring in the Great Barrier Reef: An Overview of Methods Based on Ocean Colour Satellite Data. *Remote Sensing*, **7 (10)**, 12 909–12 941, doi:10.3390/rs71012909, URL https://www.mdpi.com/2072-4292/7/10/12909.

Dykstra, S. L., G. Ricche, G. Marmorino, and A. E. Yankovsky, 2024: Forcing conditions of cross-shelf plumes on a wide continental shelf, Winyah Bay, South Atlantic Bight. *Remote Sensing of Environment*, **311**, 114 279, doi:10.1016/j.rse.2024.114279, URL https://linkinghub.elsevier.com/retrieve/pii/S0034425724002979.

Fennel, K., J. Wilkin, J. Levin, J. Moisan, J. O'Reilly, and D. Haidvogel, 2006: Nitrogen cycling in the Middle Atlantic Bight: Results from a three-dimensional model and implications for the North Atlantic nitrogen budget. *Global Biogeochemical Cycles*, **20 (3)**, 2005GB002 456, doi:10.1029/2005GB002456, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2005GB002456.

Fiedler, P. C., and R. M. Laurs, 1990: Variability of the Columbia River plume observed in visible and infrared satellite imagery. *International Journal of Remote Sensing*, **11 (6)**, 999–1010, doi:10.1080/01431169008955072, URL https://www.tandfonline.com/doi/full/10.1080/01431169008955072.

Fong, D. A., and W. R. Geyer, 2001: Response of a river plume during an upwelling favorable wind event. *Journal of Geophysical Research: Oceans*, **106 (C1)**, 1067–1084, doi:10.1029/2000JC900134, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2000JC900134.

Frey, D., and A. Osadchiev, 2021: Large River Plumes Detection by Satellite Altimetry: Case Study of the Ob–Yenisei Plume. *Remote Sensing*, **13 (24)**, 5014, doi:10.3390/rs13245014, URL https://www.mdpi.com/2072-4292/13/24/5014.

Friedland, R., T. Neumann, S. Piehl, H. Radtke, and G. Schernewski, 2025: Characterization of river plume dynamics for a better water quality management. *Frontiers in Marine Science*, **12**, 1617 660, doi:10.3389/fmars.2025.1617660, URL https://www.frontiersin.org/articles/10.3389/fmars.2025.1617660/full.

Gong, W., L. Chen, Z. Chen, and H. Zhang, 2019: Plume-to-plume interactions in the Pearl River Delta in winter. *Ocean & Coastal Management*, **175**, 110–126, doi:10.1016/j.ocecoaman.2019.04.001, URL https://linkinghub.elsevier.com/retrieve/pii/S0964569118308494.

Han, A., M. Dai, S.-J. Kao, J. Gan, Q. Li, L. Wang, W. Zhai, and L. Wang, 2012: Nutrient dynamics and biological consumption in a large continental shelf system under the influence of both a river plume and coastal upwelling. *Limnology and Oceanography*, **57 (2)**, 486–502, doi:10.4319/lo.2012.57.2.0486, URL https://aslopubs.onlinelibrary.wiley.com/doi/10.4319/lo.2012.57.2.0486.

Hetland, R. D., and D. G. MacDonald, 2008: Spreading in the near-field Merrimack River plume. *Ocean Modelling*, **21 (1-2)**, 12–21, doi:10.1016/j.ocemod.2007.11.001, URL https://linkinghub.elsevier.com/retrieve/pii/S1463500307001345.

Hickey, B. M., and Coauthors, 2010: River Influences on Shelf Ecosystems: Introduction and synthesis. *Journal of Geophysical Research: Oceans*, **115 (C2)**, 2009JC005 452, doi:10.1029/2009JC005452, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2009JC005452.

Hockensmith, B., 2004: Flow and Salinity Characteristics of the Santee River Estuary, South Carolina. SCDNR Water Resources Report 35. URL https://www.dnr.sc.gov/water/img/Report35.pdf.

Horner-Devine, A. R., R. D. Hetland, and D. G. MacDonald, 2015: Mixing and Transport in Coastal River Plumes. *Annual Review of Fluid Mechanics*, **47 (1)**, 569–594, doi:10.1146/annurev-fluid-010313-141408, URL https://www.annualreviews.org/doi/10.1146/annurev-fluid-010313-141408.

Horner-Devine, A. R., D. A. Jay, P. M. Orton, and E. Y. Spahn, 2009: A conceptual model of the strongly tidal Columbia River plume. *Journal of Marine Systems*, **78 (3)**, 460–475, doi:10.1016/j.jmarsys.2008.11.025, URL https://linkinghub.elsevier.com/retrieve/pii/S0924796309000785.

Kastner, S. E., A. R. Horner-Devine, J. M. Thomson, and S. N. Giddings, 2023: Observations of River Plume Mixing in the Surf Zone. *Journal of Physical Oceanography*, **53 (3)**, 959–977, doi:10.1175/JPO-D-21-0286.1, URL https://journals.ametsoc.org/view/journals/phoc/53/3/JPO-D-21-0286.1.xml.

Kim, Y. H., and G. Voulgaris, 2008: Lateral circulation and suspended sediment transport in a curved estuarine channel: Winyah Bay, SC, USA. *Journal of Geophysical Research: Oceans*, **113 (C9)**, 2007JC004 509, doi:10.1029/2007JC004509, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2007JC004509.

Li, D., and Coauthors, 2024: Hypoxia Triggered by Expanding River Plume on the East China Sea Inner Shelf During Flood Years. *Journal of Geophysical Research: Oceans*, **129 (8)**, e2024JC021 299, doi:10.1029/2024JC021299, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JC021299.

MacDonald, D. G., and W. R. Geyer, 2004: Turbulent energy production and entrainment at a highly stratified estuarine front. *Journal of Geophysical Research: Oceans*, **109 (C5)**, 2003JC002 094, doi:10.1029/2003JC002094, URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2003JC002094.

Ortega, E., I. Reister, S. Danielson, and A. Aguilar-Islas, 2025: Surface macro- and micro-nutrients within the Copper River plume region respond to along-shore winds. *Marine Chemistry*, **270**, 104 508, doi:10.1016/j.marchem.2025.104508, URL https://linkinghub.elsevier.com/retrieve/pii/S0304420325000234.

Osadchiev, A., and R. Sedakov, 2019: Spreading dynamics of small river plumes off the northeastern coast of the Black Sea observed by Landsat 8 and Sentinel-2. *Remote Sensing of Environment*, **221**, 522–533, doi:https://doi.org/10.1016/j.rse.2018.11.043, URL https://www.sciencedirect.com/science/article/pii/S0034425718305522.

Papageorgiou, C. T., 2023: RoboCat: An Autonomous Surface Vehicle (ASV) for Near-Surface Hydrodynamic Measurements in Buoyant Plumes. Design, Development, and the Winyah Bay Plume Case Study. M.S. thesis, University of South Carolina, United States – South Carolina, URL https://go.openathens.net/redirector/coastal.edu?url=https://www.proquest.com/dissertations-theses/robocat-autonomous-surface-vehicle-asv-near/docview/3043570534/se-2?accountid=26722, iSBN: 9798382296265 Publication Title: ProQuest Dissertations and Theses 30690018.

Patchineelam, S. M., 1999: Fine-grained sediment dynamics and budget: Winyah Bay estuary, South Carolina. Ph.D., University of South Carolina, United States – South Carolina, URL https://go.openathens.net/redirector/coastal.edu?url=https://www.proquest.com/dissertations-theses/fine-grained-sediment-dynamics-budget-winyah-bay/docview/304537630/se-2?accountid=26722, iSBN: 978-0-599-40527-1 Publication Title: ProQuest Dissertations and Theses 9939206.

Rabalais, N. N., R. E. Turner, and D. Scavia, 2002: Beyond Science into Policy: Gulf of Mexico Hypoxia and the Mississippi River. *BioScience*, **52 (2)**, 129, doi:10.1641/0006-3568(2002)052[0129:BSIPGO]2.0.CO;2, URL https://academic.oup.com/bioscience/article/52/2/129-142/341059.

Saldías, G. S., M. Sobarzo, J. Largier, C. Moffat, and R. Letelier, 2012: Seasonal variability of turbid river plumes off central Chile based on high-resolution MODIS imagery. *Remote Sensing of Environment*, **123**, 220–233, doi:https://doi.org/10.1016/j.rse.2012.03.010, URL https://www.sciencedirect.com/science/article/pii/S0034425712001290.

Small, J., 2025: Supplementary satellite imagery animations for thesis: Winyah bay and santee river plumes, 2025. Zenodo, URL https://zenodo.org/records/18164658, dataset (GIF animations), doi:10.5281/zenodo.18164658.

Thomas, A., and R. Weatherbee, 2006: Satellite-measured temporal variability of the Columbia River plume. *Remote Sensing of Environment*, **100 (2)**, 167–178, doi:10.1016/j.rse.2005.10.018, URL https://linkinghub.elsevier.com/retrieve/pii/S0034425705003482.

Warrick, J., and Coauthors, 2007: River plume patterns and dynamics within the Southern California Bight. *Continental Shelf Research*, **27 (19)**, 2427–2448, doi:10.1016/j.csr.2007.06.015, URL https://linkinghub.elsevier.com/retrieve/pii/S0278434307001720.

Warrick, J. A., and K. L. Farnsworth, 2017: Coastal river plumes: Collisions and coalescence. *Progress in Oceanography*, **151**, 245–260, doi:10.1016/j.pocean.2016.11.008, URL https://linkinghub.elsevier.com/retrieve/pii/S0079661116300532.

Wei, S., D. Fu, D. Wang, G. Yu, Y. Luo, and H. Xu, 2023: Tracer study of CDOM for nitrogen and phosphorus pollution in the offshore Leizhou Peninsula, China. *Ecological Indicators*, **148**, 110 019, doi:10.1016/j.ecolind.2023.110019, URL https://linkinghub.elsevier.com/retrieve/pii/S1470160X23001619.

Yankovsky, A. E., D. B. Fribance, D. Cahl, and G. Voulgaris, 2022: Offshore Spreading of a Supercritical Plume Under Upwelling Wind Forcing: A Case Study of the Winyah Bay Outflow. *Frontiers in Marine Science*, **8**, 785 967, doi:10.3389/fmars.2021.785967, URL https://www.frontiersin.org/articles/10.3389/fmars.2021.785967/full.

Zhong, Y., G. Yu, D. Fu, F. Chen, Y. Luo, and R. Deng, 2025: Optical properties of CDOM and assessing eutrophication by remote sensing of CDOM in the Zhanjiang Bay, China. *Frontiers in Marine Science*, **12**, 1527 200, doi:10.3389/fmars.2025.1527200, URL https://www.frontiersin.org/articles/10.3389/fmars.2025.1527200/full.